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Mechatronics Engineering | Robotics, Mechanical Design & Biomechanics

Final-year Mechatronics Engineering student at the University of São Paulo (EESC-USP).

I am a Mechatronics Engineering student at EESC–USP with academic and research experience at the University of Twente. My projects span robotic mechanisms, assistive devices, biomechanical analysis and manufacturing automation.

Across these projects, I have worked on mechanical design and CAD, modelling and simulation, prototype development, and experimental data analysis. I am particularly interested in wearable and assistive robotics, and in how mechanical design, control and experimentation come together in a working system.

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Selected projects

Project	Year	Team	My role
Hip exoskeleton joint-moment research	2026	4	Data collection; Subject modeling; Inverse dynamics analysis pipeline
HydroBot - EMG-controlled robotic arm	2025	6	Mechanical design; Dynamic model
Spider Bot - bridge cable inspection	2026	2	Mechanical design
Cookbot - Robotic Arm design	2025	5	Kinematic design; Singularity analysis
Joystick-controlled arm	2025	5	Mechanical design, actuator sizing
Drone assembly line - Design for Assembly	2026	4	Workstations framework; frame redesign

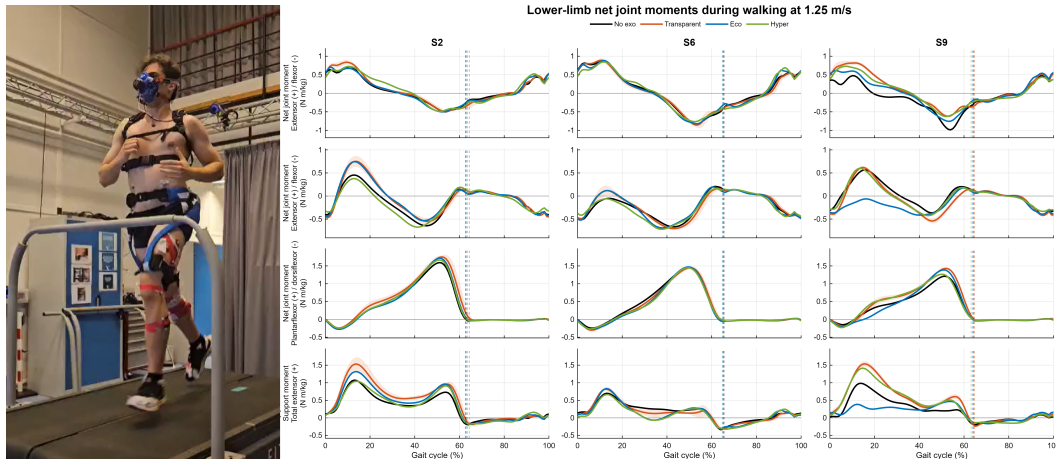
Hip Exoskeleton Assistance: Lower-Limb Joint Mechanics

Experimental gait analysis and inverse dynamics of the Hypershell X Pro during treadmill walking
Project supervised by dr.ir. Mohamed Irfan Refai, with guidance from prof.dr.ir. Edwin van Asseldonk

Where University of Twente, Biomechanical Engineering • **When** 2026 • **Team** 4-person data collection team; individual analysis • **Tools** MATLAB, OpenSim 4.5, Qualisys, Bertec instrumented treadmill

My contribution

Contributed to shared experimental data collection and individually developed the processing and analysis pipeline, performed quality control, interpreted the results and authored the report.



Left: experimental setup. Right: participant-level net joint and support moment's waveforms across four conditions during walking at 1.25 m/s ($n = 3$).

Objective

Investigate how hip-exoskeleton assistance affects net joint moments at the hip, knee and ankle. The final analysis covered three of nine recorded participants during walking at 1.25 m/s, across No-exo, Transparent, Eco and Hyper conditions. Transparent provided a device-worn reference for active-assistance comparisons.

Approach

- Developed a MATLAB processing pipeline linking motion capture and ground-reaction forces to OpenSim model scaling, inverse kinematics and inverse dynamics.
- Applied a consistent gait2392 model and marker-weighting strategy across conditions, with gait-event detection and cycle-level quality control.
- Compared joint-moment waveforms, positive and negative angular impulses, and stance support impulse within each participant, without group-level inferential statistics.

Data quality and processing

Addressed signal filtering, kinetic–kinematic alignment and model-scaling constraints. An analogue-acquisition buffer limitation restricted the valid synchronized window to 6 s per condition, yielding four to five retained gait cycles for each participant.

Key findings

Relative to Transparent, both active modes reduced positive ankle plantarflexor angular impulse in all three participants (by 3.6–17.8%) and increased the magnitude of negative hip angular impulse. Positive hip impulse showed no consistent direction of change. Stance support impulse decreased in every active-mode comparison, with participant-specific joint contributions.

Interpretation and limitations

These findings describe three individual cases rather than a population effect. Unmeasured human–device interface forces and unsuitable EMG recordings prevented separation of biological and device contributions at the hip. Short analysis windows, elevated pelvic residual forces and sensitivity of knee estimates further limited interpretation.

HydroBot — Assistive Robotic Manipulator

Mechanical design, fabrication and kinematic modelling of an EMG-controlled arm for plant watering

Where University of Twente, BioRobotics • When Q3 2025 • Team 6 students • Tools Fusion 360, Python, 3D printing, laser cutting

My contribution

Co-designed the mechanical structure and mechanisms with one teammate and together fabricated, assembled and adjusted the prototype. Developed the kinematic model, contributed to its Python implementation and supported the team's dynamic modelling. The final model was reviewed jointly before system integration.



Physical Prototype of the Robot

Objective

Develop an EMG controlled three-degree-of-freedom manipulator for plant watering. The project explored an assistive application for people with limited upper-limb strength.

Mechanical design and fabrication

- Used a four-bar linkage to actuate the second arm joint from the support platform, reducing the motor mass carried by the arm.
- Integrated geared transmissions and a custom rotating-base bearing into a structure manufactured with laser cutting and 3D printing.
- Refined component fits during assembly and used printed parts where plywood geometry or strength proved inadequate.

Kinematic modelling

Formulated the model relating end-effector motion to joint velocities through differential inverse kinematics. Jacobian-based calculations provided the link between the desired motion and the joint references used by the team's control framework.

Outcome and limitations

The team built and demonstrated an EMG-controlled prototype that could position the end effector above pots of different heights, with manual water release. Further development was needed in structural robustness and motion control before reliable watering could be demonstrated.

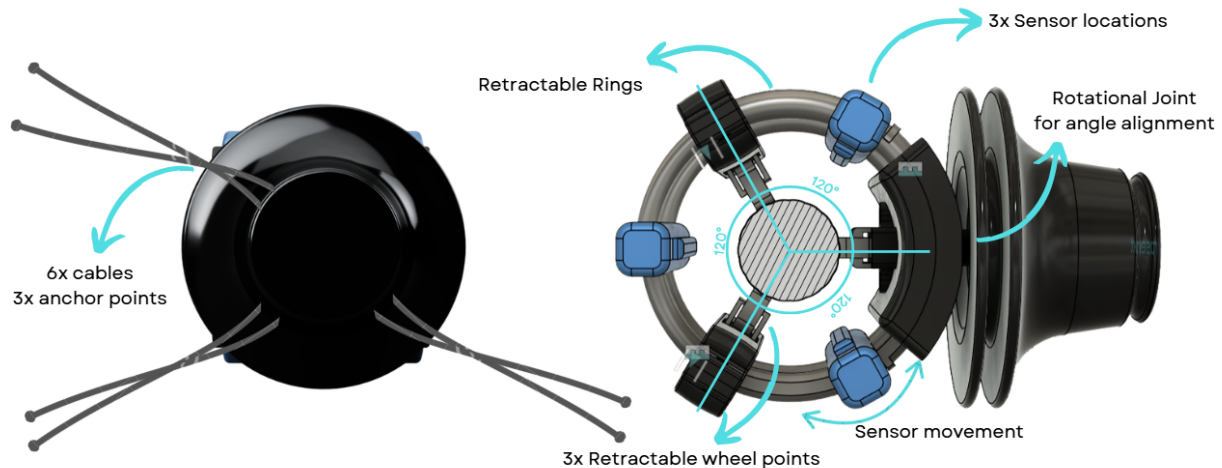
Spider Bot - Bridge Inspection Robot Concept

Concept development for stay-cable inspection, in collaboration with an external company

Where University of Twente, Robotics Innovation Tournament • When Q4 2025–Jan. 2026 • Team 2 students • Tools Fusion 360, concept development, bridge typology research

My contribution

Developed the robot's conceptual mechanical architecture and created the complete CAD model. Jointly researched bridge configurations to support concept selection. My teammate carried out the economic assessment and the mathematical modelling robot motion and stability. Final design got first place in the tournament.



Final CAD of the concept assembly

Objective

Propose a robotic system for inspecting stay cables on existing bridges. The concept needed to address access to multiple cables in different angles and be non-invasive to the bridges.

Research informing concept selection

A joint review of bridge configurations and dimensions helped relate the different robot architecture ideas to their operating environment. The analysis, along with the project requirements and stakeholder feedback helped the decision of the Cable-Driven Parallel Robot concept (the "Spider Bot").

Conceptual mechanical architecture

- Two-part body: an upper section anchored to the six actuation cables and a lower section clamping the stay cable, making the two free to rotate relative to each other so the robot can align with cables at varying inclination.
- Clamping Mechanism: Two-part rings that close around the cable and lock into each other. Retractable wheel modules for cable engagement, diameter adaptation and travel along the cable.
- Allocated space for inspection modules, batteries and onboard electronics within the overall assembly.

Outcome and development stage

Produced a CAD representation of the proposed system and its operating principle. No hardware was built and detailed mechanism design, structural verification and physical validation remained future development steps. The final concept and idea pitch was presented to the stakeholders of the partner company and was awarded first place in the Robotics Innovation Tournament course.

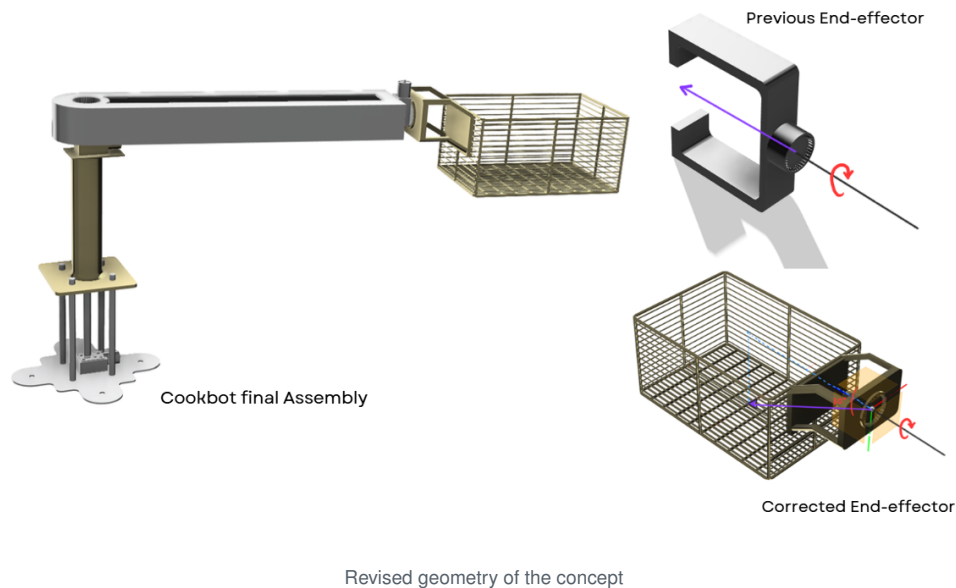
Cookbot - Singularity Analysis and CAD Revision

A PRR manipulator concept for automated frying

Where EESC-USP, Robotic Systems Control (SEM0591) • **When** First semester 2025 • **Team** 5 students • **Tools** Fusion 360, Python/SymPy analysis tools

My contribution

Investigated the singularity in the existing robot geometry, proposed and implemented a 30° end-effector modification in Fusion 360. Used a teammate's symbolic-analysis code to check the revised model and help define the task trajectory. Helped to create the dynamic model of the arm.



Problem

The inherited manipulator had one prismatic and two revolute joints. The rotation of the final joint changed tool orientation without changing the position of the selected end-effector point which produced a persistent rank deficiency in the position Jacobian.

Geometric solution

Proposed a 30° inclination of the end effector to move the point of interest away from the final rotation axis. Updated the existing CAD and used symbolic kinematic analysis to check that the revised geometry removed this persistent degeneracy in the position model.

Trajectory definition

Worked with the team to define the lifting, lowering, transfer and tipping sequence for the frying task. Analyzed singular configurations encountered along this sequence and contributed to the joint-space trajectory definition.

Outcome and scope

Delivered revised CAD geometry and an analysis supporting the change, alongside contributions to the team's trajectory study.

The work demonstrated the proposed correction at model level. Controller development and tuning were handled by other team members.

Joystick-Controlled Robotic Arm

Mechanical CAD and actuator sizing for a manufactured and tested student prototype

Where EESC-USP, Mechatronics Engineering Projects I • **When** First semester 2025 • **Team** 5 students • **Tools** Solid Edge, actuator sizing, 3D printing

My contribution

Created the mechanical CAD in Solid Edge and performed the motor-sizing calculations. The team manufactured, assembled and tested the joystick-controlled prototype.



Physical Prototype of the Robot

Objective

Design and build a robotic arm that could be operated through a joystick and lift a test load of 200 g. The project connected mechanical geometry and actuator selection with a physical prototype.

Mechanical development

Developed the arm's CAD geometry and assessed the motor requirements for the joints. The structure was produced using 3D printing and assembled with the selected actuators.

Prototype testing

All joints moved and the arm lifted the test load. However, operators could not reliably reach or maintain the desired joint positions, since continuous-rotation servos without joint-position feedback were used, and no position-control loop was implemented in the system.

Project takeaways

The tests exposed a gap between load-carrying capability and positioning performance. A revised design would apply closed-loop control using position feedback to make sure the wanted results were accomplished.

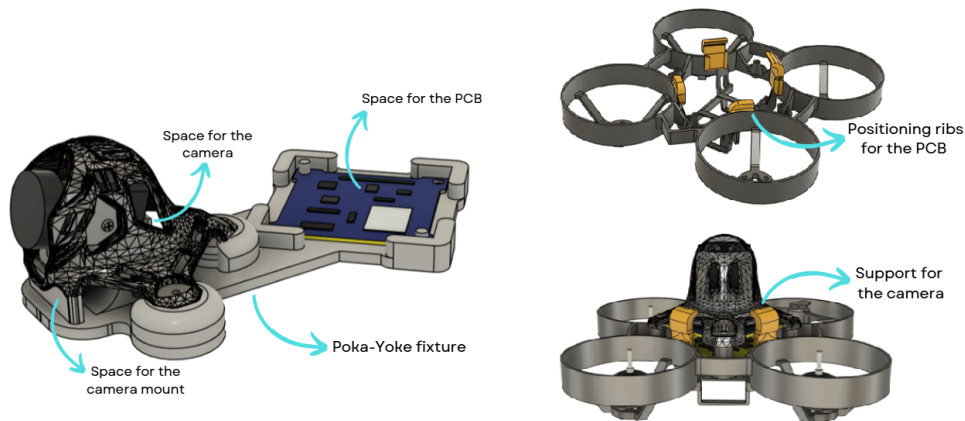
Drone Assembly — Workstation and Product Design

Manual workstation planning and CAD changes for a proposed hybrid production line

Where University of Twente, Automated Production Systems • **When** Q4 2025–Jan. 2026 • **Team** 4 students • **Tools** Process planning, time study, fixture design, CAD

My contribution

Defined the processes, selected components and estimated operation times for manual workstations 2 and 3. Designed the camera assembly fixture and modified the drone chassis CAD to facilitate assembly. Participated in drone disassembly, reassembly and timing. A teammate implemented the workstations in Visual Components.



Left: Camera assembly fixture. Right: Revised drone chassis

Objective

Develop a hybrid production-line concept combining manual assembly with robotic operations to make a drone assembly process easier and more scalable. Observation and timing of the original assembly process provided a baseline for proposed improvements.

Manual workstation planning

- **Workstation 2 — camera sub-assembly:** Defined the process and designed a fixture to support the camera housing, freeing the operator's hands for insertion and screw alignment.
- **Workstation 3 — PCB and camera placement:** Defined the component-placement sequence and estimated the required operation times for integration into the main drone frame.

Design for assembly

Added locating ribs to the chassis to constrain the drone's PCB and guide features to align the camera module. These changes were intended to reduce manual alignment difficulties and support repeatable positioning for subsequent fastening operations.

Outcome and evidence

Delivered workstation process definitions, component selections, operation-time estimates, a fixture concept and revised chassis CAD. The original assembly process was physically examined and timed but the proposed fixture and chassis changes were not manufactured or tested. Estimated improvement combined the team's proposed changes.